

Mathis BENCHIKH

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PORTFOLIO : [mtsben.github.io](https://github.com/mtsben) | LinkedIn : [Mathis Benchikh](#)



PROFILE: Graduate engineering student pursuing the Arts et Métiers–Cranfield double degree with the MSc in Aerospace Vehicle Design, seeking a **4–6-month final-year internship starting in May 2027**. I aim to contribute to industrial aerospace R&D projects while deepening my expertise in design, numerical simulation and guidance and control.

EDUCATION

- Cranfield University**, Cranfield, United Kingdom Sep 2026 – Sep 2027
MSc Aerospace Vehicle Design — Double degree with Arts et Métiers (in progress)
- Coursework: aircraft systems, aerodynamics, stability and control, composite structures, certification and safety.
 - Project: hydrogen tank design for the Cobalt Blue 3 project, sponsored by **Safran**
- Arts et Métiers**, Paris / Châlons-en-Champagne, France Sep 2024 – Jun 2027
Generalist Engineering Degree — Mechanical and Industrial Engineering (in progress)
- **Ranking: National top 9%** (110 / 1,200 students); average grade: 15.8/20.
 - Coursework: fluid and structural mechanics, thermodynamics, electrical engineering and numerical simulation.
 - Student association: Head of Theory and Integration — AMSPACE (Oct 2025 – Feb 2026).
- Scientific Preparatory Classes, Lycée Fauriel**, Saint-Étienne, France Sep 2022 – Jun 2024
- MPSI–MP*: Mathematics, Physics and Engineering Sciences — Top 8 of 47 in MP*.
 - Intensive preparation for competitive entrance examinations to France’s leading engineering schools.

PROFESSIONAL EXPERIENCE

- CNRS@CREATE**, Singapore Jun 2026 – Aug 2026
Research Internship — Trajectory Optimisation for Autonomous Drones
- Co-author: contributed to writing a research paper** submitted to the Journal of Control, Automation and Electrical Systems, **currently under review.**
- Co-developed a variational planner (Euler–Lagrange) and a collision-avoidance correction algorithm with the PIMM laboratory in a high-level international research environment.
 - Integrated the perception–guidance–control pipeline in Python: closed-loop trajectory tracking at approximately 20 Hz, validated on an experimental rover platform.
 - Team validation: 40/40 goals reached without collisions across two repeated scenarios; simulation–experiment comparison and analysis of tracking accuracy and safety margins.
- Emitech Group — LEFAE**, Saint-Chamond, France Jun 2025 – Jul 2025
Shop-floor Internship — Aerospace Testing and Qualification
- Participated in qualification tests for Airbus A350F cargo-door seals: specimen preparation, stiffness testing and leak testing under pressure.
 - Instrumentation and data acquisition: thermocouples, pressure and displacement sensors; analysis of force–displacement curves.
 - Participated in four lithium-ion battery thermal-runaway propagation tests; prepared and monitored tests in accordance with safety and traceability protocols.
- ACADOMIA**, Paris / Châlons-en-Champagne, France Jan 2025 – Jun 2026
Private Tutor — Mathematics, Physics and Chemistry
- Provided tailored support to 10 secondary-school students: adapted explanations to individual needs and monitored progress.
 - Developed students’ scientific reasoning, study methods and independent learning skills.

SKILLS AND INTERESTS

Technical skills: CAD: CATIA V5, Fusion 360, 3DEXPERIENCE | Simulation: STAR-CCM+, Abaqus | Programming: Python, MATLAB/Simulink.

Personal strengths: Rigour, Communication, Stress management, Adaptability, Resilience, Teamwork.

Languages: French (native), English (C1), Spanish (B1), Tamazight.

Interests: Hiking, travelling, football (R2 level), kickboxing, amateur esports.